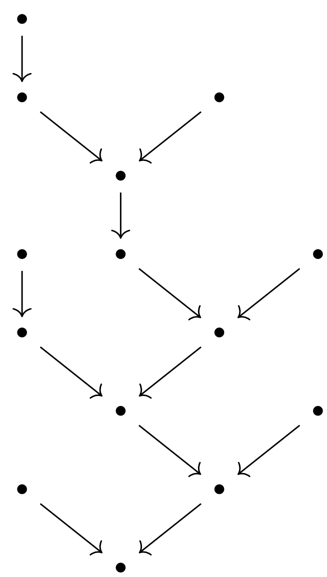
$$\dfrac{\exists x\,Px \quad \dfrac{\dfrac{\dfrac{[\forall x\,\exists y\,\neg Rxy]}{\exists y\,\neg Ray} \quad \dfrac{\dfrac{\dfrac{\dfrac{\forall x\,(Pa \to \forall y\,Ray)}{Pa \to \forall y\,Ray} \quad [Pa]}{\forall y\,Ray}}{Rab} \quad [\neg Rab]}{\neg\forall x\,\exists y\,\neg Rxy}}{\neg\forall x\,\exists y\,\neg Rxy} \quad [\forall x\,\exists y\,\neg Rxy]}{\neg\forall x\,\exists y\,\neg Rxy}}{\neg\forall x\,\exists y\,\neg Rxy}$$

Clearly, this has the structure of a tree, with the conclusion ($\neg\forall x\,\exists y\,\neg Rxy$) as its root. If we forget about the contents of the different steps and just draw their structure as a graph, this is what we get:



We'll explain how to make this more precise, by setting out a mathematical description of the tree structure itself, in terms of nodes and edges, and then adding a *labelling* in terms of formulas. To a large degree, however, this is included only for thoroughness: it doesn't make the presentation substantially clearer than the informal presentation in TLM, and it isn't required for the key proofs in Chapters 2 and 3. (In fact, when we state the inference rules, we'll do so graphically, in the style of TLM, rather than defining explicit operations on trees.) Thus, you should free to skim the details.

A **partially ordered set** is a pair $\langle S, \preceq \rangle$ consisting of a set S and a relation $\preceq \subseteq S \times S$ where $\preceq$ is transitive and antisymmetric. (For simplicity, we write $x \preceq y$ instead of $\langle x, y \rangle \in \preceq$, $x \npreceq y$ for the negation of $x \preceq y$, and $x \prec y$ where $x \preceq y$ and $y \npreceq x$.) A **linearly ordered set** is a partially ordered set where in addition, for all $x, y \in S$, $x \preceq y$ or $y \preceq x$ is a partial order. When we describe a subset S' of a given partially ordered set $(S, \preceq)$ as linearly ordered, the relevant ordering relation is the restriction of $\preceq$ to the members of S'.

A **tree** is a partially ordered set $\mathscr{T} = \langle S, \preceq \rangle$ with the following additional properties: (iii) S is finite; (iv) there exists a unique $x \in S$ such that, for all $y \in S$, $x \preceq y$; and (v) for all $y \in S$, the subset $R(y) = \{x \in S : x \preceq y\}$ is linearly ordered. (In graph theory, a 'tree' in our sense would be called a *finite rooted directed tree.*)

We term the elements of S **nodes**, and the distinguished minimum element mentioned in clause (iv) the **root**. We term a node $x \in S$ a **leaf** if there is no $y \in S$ such that $x \prec y$. A **branch** is a linearly ordered subset $B \subseteq S$ such that B is maximal, i.e. for any $x \in S$ such that $x \notin B$, $B \cup \{x\}$ is not linearly ordered. We define the length of a branch B, $\mathrm{lh}(B)$, as the number of elements of B, and the length of a tree $\mathscr{T}$, $\mathrm{lh}(\mathscr{T})$, as the length of its longest branch. Given a node $x \in S$, we say that $y \in S$ is a **successor** of x if $x \prec y$ and an **immediate successor** of x if $x \prec y$ and there exists no $z \in S$ such that $x \prec z$ and $z \prec y$. (The length function will play an important role, for it will permit us to prove results by induction on length of proofs, much as we proved results by induction on complexity of formulae.)

A **labelled tree**, for our purposes, is a pair $\pi = \langle \mathscr{T}, f \rangle$ where $\mathscr{T} = \langle S, \preceq \rangle$ is a tree and $f : S \longrightarrow \mathsf{Sent}$ assigns to each node a sentence of $\mathscr{L}_=$. Much as we were not interested in *all* strings of $\mathscr{L}_=$, we are not interested in all trees that are labelled by sentences in $\mathscr{L}_=$, but only some: the $\mathscr{L}_=$**-proofs**.

A proof is a labelled tree meeting certain additional requirements, which we define recursively. These requirements represent the inference rules of $\mathrm{ND}_=$. We present the definition below; as noted above, it is designed for readability, and thus we freely use diagrams to illustrate how a proof relates to its subproofs. (A subproof of a proof $\pi = \langle \mathscr{T}, f \rangle = \langle \langle S, \preceq \rangle, f \rangle$ is a labelled tree $\pi_1 = \langle \mathscr{T}_1, f_1 \rangle = \langle \langle S_1, \preceq_1 \rangle, f_1 \rangle$ which is also a proof, where $S_1 \subseteq S$ and $\preceq_1$ and f_1 are the restrictions of $\prec$ and f, respectively, to S_1.)

In general, we shall use π, ρ, and variants for proofs. If π is a proof, then we term the sentence φ labelling its root node its **conclusion**, and we say that π is a **proof of** φ. We term the set of all sentences labelling a leaf of π the **assumptions** of π. An assumption can be either **open** or **closed** (or **discharged**); we define the set of **open assumptions** of a proof π, $\mathsf{As}(\pi)$, in the course of the recursive definition below.

Definition 1.3.1 (Proofs and Open Assumptions). We start with the base case, where a proof consists of a single node. There are two versions, one of which applies to all the proof systems, and one of which applies only to $\mathrm{ND}_=$.

(Assumption) A labelled tree consisting of a single node, labelled with the sentence φ, constitutes a proof π of φ of the form

$$\varphi$$

with $\mathsf{As}(\pi) = \{\varphi\}$.

The special case for $\mathrm{ND}_=$ is (=Intro), which allows us to introduce a statement of the form $\mathbf{c} = \mathbf{c}$ without any additional assumptions:

(=Intro) A labelled tree consisting of a single node, labelled with the sentence $\mathbf{c} = \mathbf{c}$, constitutes a proof π of $\mathbf{c} = \mathbf{c}$ of the form

$$\mathbf{c} = \mathbf{c}$$

with $\mathsf{As}(\pi) = \varnothing$.

We now cover the rules for ND_1 (which also apply, of course, to ND_2 and $\mathrm{ND}_=$). To aid clarity, we introduce the following notational conventions: names for proofs are boxed, and we add a superscripted formula or formulae in brackets next to the proof to indicate any assumptions discharged. (Neither the box nor the brackets are really part of the proof, and the discharged assumptions are officially distinguished from open ones only through the bookkeeping function As.)

($\wedge$Intro) Where π_1 is a proof of φ_1 and π_2 is a proof of φ_2, the labelled tree π with the following structure constitutes a proof of $\varphi_1 \wedge \varphi_2$:

$$\dfrac{\begin{matrix}\boxed{\pi_1} \\ \vdots \\ \varphi_1\end{matrix} \qquad \begin{matrix}\boxed{\pi_2} \\ \vdots \\ \varphi_2\end{matrix}}{\varphi_1 \wedge \varphi_2}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)$.

($\wedge$Elim1) Where π_1 is a proof of $\varphi_1 \wedge \varphi_2$, the labelled tree π with the following structure constitutes a proof of φ_1:

$$\dfrac{\begin{matrix}\boxed{\pi_1} \\ \vdots \\ \varphi_1 \wedge \varphi_2\end{matrix}}{\varphi_1}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

($\wedge$Elim2) Where π_1 is a proof of $\varphi_1 \wedge \varphi_2$, the labelled tree π with the following structure constitutes a proof of φ_2:

$$\cfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots \\ \varphi_1 \wedge \varphi_2\end{array}}{\varphi_2}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

($\vee$Intro1) Where π_1 is a proof of φ_1, the labelled tree π with the following structure constitutes a proof of $\varphi_1 \vee \varphi_2$:

$$\cfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots \\ \varphi_1\end{array}}{\varphi_1 \vee \varphi_2}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

($\vee$Intro2) Where π_1 is a proof of φ_2, the labelled tree π with the following structure constitutes a proof of $\varphi_1 \vee \varphi_2$:

$$\cfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots \\ \varphi_2\end{array}}{\varphi_1 \vee \varphi_2}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

($\vee$Elim) Where π_1 is a proof of φ, π_2 is a proof of φ, and π_3 is a proof of $\psi_1 \vee \psi_2$, the labelled tree π with the following structure constitutes a proof of φ:

$$\cfrac{\begin{array}{c}\boxed{\pi_1}^{[\psi_1]}\\ \vdots \\ \varphi\end{array} \qquad \begin{array}{c}\boxed{\pi_2}^{[\psi_2]}\\ \vdots \\ \varphi\end{array} \qquad \begin{array}{c}\boxed{\pi_3}\\ \vdots \\ \psi_1 \vee \psi_2\end{array}}{\varphi}$$

and $\mathsf{As}(\pi) = (\mathsf{As}(\pi_1) \setminus \{\psi_1\}) \cup (\mathsf{As}(\pi_1) \setminus \{\psi_2\}) \cup \mathsf{As}(\pi_3)$.

(→Intro) Where π_1 is a proof of φ, the labelled tree π with the following structure constitutes a proof of $\psi \to \varphi$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}^{[\psi]}\\ \vdots \\ \varphi\end{array}}{\psi \to \varphi}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \setminus \{\psi\}$.

(→Elim) Where π_1 is a proof of ψ and π_2 is a proof of $\psi \to \varphi$, the labelled tree π with the following structure constitutes a proof of φ:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots \\ \psi\end{array} \qquad \begin{array}{c}\boxed{\pi_2}\\ \vdots \\ \psi \to \varphi\end{array}}{\varphi}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)$.

(¬Intro) Where π_1 is a proof of ψ and π_2 is a proof of $\neg\psi$, the the labelled tree π with the following structure constitutes a proof of $\neg\varphi$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}^{[\varphi]}\\ \vdots \\ \psi\end{array} \qquad \begin{array}{c}\boxed{\pi_2}^{[\varphi]}\\ \vdots \\ \neg\psi\end{array}}{\neg\varphi}$$

and $\mathsf{As}(\pi) = (\mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)) \setminus \{\varphi\}$.

(¬Elim) Where π_1 is a proof of ψ and π_2 is a proof of $\neg\psi$, the the labelled tree π with the following structure constitutes a proof of φ:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}^{[\neg\varphi]}\\ \vdots \\ \psi\end{array} \qquad \begin{array}{c}\boxed{\pi_2}^{[\neg\varphi]}\\ \vdots \\ \neg\psi\end{array}}{\varphi}$$

and $\mathsf{As}(\pi) = (\mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)) \setminus \{\neg\varphi\}$.

(↔Intro) Where π_1 is a proof of φ_2 and π_2 is a proof of φ_1, the labelled tree π with the following structure constitutes a proof of $\varphi_1 \leftrightarrow \varphi_2$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}^{[\varphi_1]}\\ \vdots\\ \varphi_2\end{array}\qquad\begin{array}{c}\boxed{\pi_2}^{[\varphi_2]}\\ \vdots\\ \varphi_1\end{array}}{\varphi_1 \leftrightarrow \varphi_2}$$

and $\mathsf{As}(\pi) = (\mathsf{As}(\pi) \setminus \{\varphi_1\}) \cup (\mathsf{As}(\pi) \setminus \{\varphi_2\})$.

(↔Elim1) Where π_1 is a proof of $\varphi_1 \leftrightarrow \varphi_2$ and π_2 is a proof of φ_1, the labelled tree π with the following structure constitutes a proof of φ_2

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \varphi_1 \leftrightarrow \varphi_2\end{array}\qquad\begin{array}{c}\boxed{\pi_2}\\ \vdots\\ \varphi_1\end{array}}{\varphi_2}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)$.

(↔Elim2) Where π_1 is a proof of f $\varphi_1 \leftrightarrow \varphi_2$ and π_2 is a proof of φ_2, the labelled tree π with the following structure constitutes a proof of φ_1

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \varphi_1 \leftrightarrow \varphi_2\end{array}\qquad\begin{array}{c}\boxed{\pi_2}\\ \vdots\\ \varphi_2\end{array}}{\varphi_1}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)$.

Now we cover the inference rules characteristic of ND_2 (which also apply to $\mathrm{ND}_=$).

(∀Intro) Where π_1 is a proof of $\varphi[{}^{c}/_{v}]$, $\mathbf{c}$ does not occur in φ, and $\mathbf{c}$ does not occur in ψ for any $\psi \in \mathsf{As}(\pi_1)$, the labelled tree π with the following structure constitutes a proof of $\forall \mathbf{v}\, \varphi$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \varphi[{}^{c}/_{v}]\end{array}}{\forall \mathbf{v}\, \varphi}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

(∀Elim) Where π_1 is a proof of $\forall\mathbf{v}\,\varphi$, the labelled tree π with the following structure constitutes a proof of $\varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \forall\mathbf{v}\,\varphi\end{array}}{\varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

(∃Intro) Where π_1 is a proof of $\varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]$, the labelled tree π with the following structure constitutes a proof of $\exists\mathbf{v}\,\varphi$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]\end{array}}{\exists\mathbf{v}\,\varphi}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1)$.

(∃Elim) Where π_1 is a proof of $\exists\mathbf{v}\,\varphi$, π_2 is a proof of ψ, $\mathbf{c}$ does not occur in $\exists\mathbf{v}\,\varphi$, and $\mathbf{c}$ does not occur in χ for any $\chi \in \mathsf{As}(\pi_2) \setminus \{\varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]\}$, the labelled tree π with the following structure constitutes a proof of ψ:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \exists\mathbf{v}\,\varphi\end{array} \qquad \begin{array}{c}\boxed{\pi_2}^{[\varphi[{}^{\mathrm{c}}/_{\mathrm{v}}]]}\\ \vdots\\ \psi\end{array}}{\psi}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup (\mathsf{As}(\pi_2) \setminus \varphi[{}^{\mathrm{c}}/_{\mathrm{v}}])$.

You should be familiar with the motivation behind the restrictions (termed **side conditions** on (∀Intro) and (∃Elim), as set out in TLM, and be able to spot proofs where they are used incorrectly.

Finally, we cover the elimination rules specific to $\mathrm{ND}_=$. (The introduction rule was already covered in the base case.)

(=Elim1) Where π_1 is a proof of $\mathbf{c}_1 = \mathbf{c}_2$ and π_2 is a proof of φ, the labelled tree π with the following structure constitutes a proof of $\varphi[{}^{\mathbf{c}_2}/_{\mathbf{c}_1}]$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \mathbf{c}_1 = \mathbf{c}_2\end{array} \qquad \begin{array}{c}\boxed{\pi_2}\\ \vdots\\ \varphi\end{array}}{\varphi[\mathbf{c}_2/\mathbf{c}_1]}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup (\mathsf{As}(\pi_2)$.

(=Elim2) Where π_1 is a proof of $\mathbf{c}_1 = \mathbf{c}_2$ and π_2 is a proof of φ, the labelled tree π with the following structure constitutes a proof of $\varphi[\mathbf{c}_1/\mathbf{c}_2]$:

$$\dfrac{\begin{array}{c}\boxed{\pi_1}\\ \vdots\\ \mathbf{c}_1 = \mathbf{c}_2\end{array} \qquad \begin{array}{c}\boxed{\pi_2}\\ \vdots\\ \varphi\end{array}}{\varphi[\mathbf{c}_1/\mathbf{c}_2]}$$

and $\mathsf{As}(\pi) = \mathsf{As}(\pi_1) \cup \mathsf{As}(\pi_2)$.

We now introduce a few more definitions, as well as the familiar turnstile notation for proofs (and some variations).

Definition 1.3.2. Where $\varphi \in \mathsf{Sent}_\bullet$ and $\Gamma \subseteq \mathsf{Sent}_\bullet$, we write $\Gamma \vdash^{\pi} \varphi$ for: π is a proof of φ with $\mathsf{As}(\pi) \subseteq \Gamma$. Where $\mathrm{lh}(\pi) = n$, we also write $\Gamma \vdash^{\pi}_{n} \varphi$; if $\mathrm{lh}(\pi) < n$, then we write $\Gamma \vdash^{\pi}_{<n} \varphi$. We write $\Gamma \vdash \varphi$ for: there exists some proof π such that $\Gamma \vdash^{\pi} \varphi$.

A **theory** is a set of sentences closed under derivability: i.e., a $\Gamma \subseteq \mathsf{Sent}_\bullet$ is a theory just in case, if $\Gamma \vdash \varphi$, then $\varphi \in \Gamma$. (Note that the other direction, that if $\varphi \in \Gamma$ then $\Gamma \vdash \varphi$ is trivial.) We say that a $\Gamma \subseteq \mathsf{Sent}_\bullet$ is comprised of the **axioms** for a theory Δ, or that Γ is an **axiomatization** of Δ, when $\Delta = \{\varphi \in \mathsf{Sent}_\bullet : \Gamma \vdash \varphi\}$.

A set $\Gamma \subseteq \mathsf{Sent}_\bullet$ is **syntactically consistent** if and only if it is not the case that, for all $\varphi \in \mathsf{Sent}_\bullet$, $\Gamma \vdash \varphi$.

We use abbreviation conventions for $\vdash$ similar to those we used for $\vDash$: thus we write $\vdash \varphi$ instead of $\varnothing \vdash \varphi$, $\psi_1, \ldots, \psi_n \vdash \varphi$ instead of $\{\psi_1, \ldots, \psi_n\} \vdash \varphi$, $\Gamma, \Delta \vdash \varphi$ for $\Gamma \cup \Delta \vdash \varphi$, and $\Gamma, \psi \vdash \varphi$ for $\Gamma \cup \{\psi\} \vdash \varphi$.

Our definition of syntactic consistency is the standard one, but there is a related property of note: a $\Gamma \subseteq \mathsf{Sent}_\bullet$ is **negation-consistent** if and only if it is not the case that, for some $\varphi \in \mathsf{Sent}_\bullet$, both $\Gamma \vdash \varphi$ and $\Gamma \vdash \neg\varphi$. (In a language without negation, our notion of consistency would make sense, whereas negation-consistency would not; it is thus more general.)

We show that, for all three of our systems, syntactic consistency and negation-consistency coincide: